ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Simultaneous Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SIMULTANEOUS EQUATIONS · CH2 · L12

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The Map

This lesson collapses Simultaneous Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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FIRST LINE TO WRITE

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FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x + 7y = 17$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q12

Solve simultaneous equations

$$2x + 7y = 17$$

Multiply the first equation by

$$10x + 35y = 85$$

Finish: $x = -2$, $y = 3$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve simultaneous equations

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.